

Mitigating Capital Allocation Risk through SEM-Informed Stratification: A Density-Based Approach to Evaluating Regional Treatment Effects

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Abstract

Geographic experimentation has emerged as a robust methodology for measuring the causal return on localized capital deployments (e.g., incremental Return on Investment), particularly as privacy regulations increasingly restrict user-level tracking. However, executing randomized geographic interventions at scale is hindered by market volatility and the inherent structural heterogeneity across regional units. Traditional "greedy" matching algorithms often fail in these environments by pairing fundamentally incompatible geographic regions based solely on superficial short-term revenue correlations. In addition, it introduces unseen structural risk, as financial stakeholders cannot easily evaluate the underlying causal drivers of these pairings. This paper proposes a multi-stage causal model¹ designed explicitly to validate experimental designs prior to capital deployment. The architecture integrates panel Structural Equation Modeling (SEM) to map market-specific causal responsiveness, weighted DBSCAN clustering to stratify regions into structurally homogeneous geographic cohorts, and synthetic capital-injection to simulate regional interventions and calculate the Minimum Detectable Effect (MDE). Evaluated on an aggregated dataset of 208 Designated Market Areas (DMAs), the researchers demonstrate that restricting matched-market algorithms to structurally validated geographic cohorts eliminates treatment effect heterogeneity bias, passes rigorous safety gates (AA tests, drift checks, correlation thresholds), and provides decision-ready MDEs before any live capital is committed.

Keywords: Geographic Experiments, Minimum Detectable Effect (MDE), Structural Equation Modeling (SEM), Causal Discovery, DBSCAN Clustering, Time-Based Regression, Matched-Market Design, Greedy Matching Bias, Experimental Validation

Introduction

Background of the Study

¹The anonymized codebase to reproduce this framework is available at: <https://anonymous.4open.science/r/Robust-Geographical-Incrementality-Testing-4E44/>

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Corporate companies employ multi-billion dollar strategic capital in extremely volatile decentralized digital platforms. Traditional based financial models assess risk within defined asset classes, whereas digital advertising is a high-velocity, high-variance empirical proxy for testing spatial capital allocation. Thus, localized marketing channels are not simply operating expenses, but separate pieces in a multi-channel capital allocation portfolio. Rules governing privacy (GDPR, cookie deprecation) have restricted user-level monitoring, leading finance and operational leaders to move to aggregated measuring strategies to validate these investments (Braun and Schwartz 2025). Regional randomized controlled trials—commonly executed as geographic experiments in digital ecosystems—are now the gold standard for calculating the incremental return on these investments. Compared to user-level A/B tests that are confounded by algorithmic selection bias and require sensitive user data, geographic experiments are robust to network spillover and privacy constraints (Vaver and Koehler 2011; Barajas and Bhamidipati 2021).

However, the significant structural variety of the spatial landscape prevents the execution of geographic experiments at scale. Standard "greedy" matching algorithms match test and control regions based on naive correlations of pre-period revenues, often matching structurally incompatible locations (Chen, Longfils, and Remy 2021). Existing stratification approaches, e.g., random "Supergeo" stratifications, do not rigorously incorporate causal priors (Shaw 2025). Such weaknesses raise variance and contradict parallel trends assumptions and finally result in inaccurate treatment effect estimates (Gong et al. 2024).

Problem Formulation

Let $\mathcal{U} = \{u_1, u_2, \dots, u_N\}$ be a collection of diverse geographic regions. The experimental design aims to divide $\mathcal{U}$ into treatment ($\mathcal{T}$) and control ($\mathcal{C}$) cohorts to assess the incremental effect of an intervention. Standard greedy matching minimizes a distance function $d(u, v)$ that depends only on a univariate pre-period goal metric Y . If Y is confounded by unobserved structural heterogeneity (e.g., unequal receptivity to localized financial interventions), it $d(u, v)$ reflects false correlations. Matching N highly diverse units unconstrained is a combinatorial optimization problem that is NP-hard and subject to structural mismatch.

In this work, the authors suggest a multi-staged causal framework that validates experimental designs prior to the deployment of the capital investment. The approach consists of: panel SEM to generate region-specific causal responsiveness priors; weighted DBSCAN to stratify regions into homogenous cohorts; and cluster-constrained greedy matching to avoid cross-type contamination. The Minimum Detectable Effect (MDE) is computed using a synthetic capital-injection technique, and structural integrity checks are performed utilizing safety gates (AA tests, drift checks and correlation diagnostics). The authors apply this to 208 US DMAs and demonstrate that causal-informed stratification removes the heterogeneity bias, allows precise MDE computation, and enables decision-ready validation prior to any live capital.

Problem Statement

- **Greedy Matching Over Structural Alignment:** Standard approaches match on pre-period revenue trends, capturing false correlations rather than structural alignment (Chen, Longfils, and Remy 2021; Callaway and Sant’Anna 2021).
- **Causally-Blind Stratification:** Advanced variance-minimization approaches (e.g., ‘Supergeo’ and PCA) mathematically balance covariates but operate as causally blind black boxes as it groups regions by revenue fluctuation magnitude rather than the cause of those fluctuations. (Gong et al. 2024; Shaw 2025).
- **No Pre-Experiment Validation:** Matched designs are only validated post-experiment with no pipeline to compute MDE against statistical thresholds (Barajas and Bhamidipati 2021).
- **Capital Deployment Risk:** Financial stakeholders cannot distinguish between true treatment effect and baseline model misspecification, resulting in unquantified financial exposure and a loss of trust in the budget allocation (Boegershausen et al. 2025).

Purpose and Objectives of the Study

The major goal of this study is to develop and evaluate a repeatable causal model of geo-experiment design that augments correlation-based matching with causal-aware stratification and enables decision-ready validation before capital deployment.

To meet this purpose, the study focuses on the following specific objectives:

- **To derive region-specific causal responsiveness priors:** Fit a panel Structural Equation Model (SEM) with fixed effects to model the causal impact of capital investment on revenue, generating responsiveness weights.
- **To stratify geographic regions into structurally homogeneous cohorts:** Use DBSCAN clustering with weights from SEM, using revenue, visitors, orders, and spend attributes to cluster regions and separate volatile outliers.
- **To develop a cluster-constrained greedy matching protocol:** Constrain Time-Based Regression (TBR) matching algorithms to work only inside DBSCAN-derived co-

horts, avoiding cross-type matches that add treatment effect heterogeneity bias.

- **To establish a validation and power analysis protocol:** Run synthetic interventions to estimate the Minimum Detectable Effect (MDE) using AA tests, drift checks, and correlation diagnostics to assure design integrity before deploying capital.
- **To empirically validate the framework:** The authors apply the proposed pipeline to 208 US Designated Market Areas (DMAs) to show that causal-informed stratification eliminates heterogeneity bias, supports precise MDE computation, and provides decision-ready validation before real capital is deployed.

Importance of the Study

This study contributes to applied computing and intelligent decision support systems in three distinct methods:

- **Algorithmic Pipeline** incorporating SEM, weighted DBSCAN and cluster-constrained greedy matching to decrease treatment effect heterogeneity (Chen and Au 2022; Shaw 2025).
- **A Decision-Support Protocol** pre-experiment validation system that calculates MDE and performs safety checks before capital deployment (Barajas, Zidar, and Bay 2020; Kerman, Wang, and Vaver 2017).
- **Empirical Benchmarking:** The authors evaluate the methodology on a dataset of 208 US Designated Market Areas (DMAs), comparing cluster-constrained to unconstrained matching in a controlled setting (Chen, Longfils, and Remy 2021; Chen et al. 2023).

Scope and Limitations

This study is limited to design validation on 208 US DMAs with a 28-day pre-test window. Limitations: outlier removal of DMAs, single pre-test window length, no post-treatment effect estimate. In future study, the researcher will extend the system to triangulated treatment effect estimation (DiD, QRF, Bayesian STS) and adaptive clustering. The exclusion of 9 outlier DMAs constrains total geographic coverage but is indicative of realistic capital allocation constraints where highly volatile and unmatchable markets are triaged and managed with separate independent operational protocols versus forced experimental pairs.

The remainder of this paper is structured as follows: related work (Section 2), methodology (Section 3), results and discussion (Section 4), conclusion (Section 5), and future works (Section 6).

Related Works

Geo-Experiments in a Privacy-First Era

With the deprecation of third-party cookies and the advent of privacy regulations, user-level measurement has become less reliable (Braun and Schwartz 2025; Liu, Wang, and Lin 2025). The industry has shifted to geo-experiments that are robust to network spillover and privacy constraints (Barajas and Bhamidipati 2021). Recent work positions

geo-experiments as the gold standard for measuring causal returns on localized capital deployments (Vaver and Koehler 2011).

Limitations of Myopic "Greedy" Matching

However, geographic experiments at scale are challenged by “greedy” matching algorithms which match test and control regions only on the similarity of pre-treatment outcomes (Chen, Longfils, and Remy 2021). This observational approach breaks parallel trends’ assumptions and inflates variance (Callaway and Sant’Anna 2021).

Limitations of “Supergeo” Designs

Supergeo design utilizes Principal Component Analysis (PCA) and Mixed-Integer Linear Programming (MILP) to reduce dimensionality and partition regions into balanced blocks (Chen et al. 2023; Shaw 2025). However, PCA is inherently a variance-maximization technique. Grouping regions by raw observational variance fails to isolate the underlying causal topology; it groups regions by how much they fluctuate, rather than why they fluctuate (Gong et al. 2024).

Deriving Causal Priors with SEM

To overcome the limitations of observational clustering, Structural Equation Modeling (SEM) has been used to derive causal relationships and develop "responsiveness priors" (Henseler 2017). This moves experimental design from guessing similarity to measuring structural responsiveness (Gong et al. 2024).

DBSCAN and Pre-Experiment Validation

Traditional geographic matching algorithms treat each region as a test or control pair, even when volatile regions are unmatchable outliers. This is done via density-based spatial clustering (DBSCAN), which removes noise so that only structurally homologous units enter the matching. Finally, a major gap remains: the lack of consistent pre-experiment validation. Recent guidelines highlight synthetic interventions, synthetic capital-injection simulations to calculate the Minimum Detectable Effect (MDE), and A/A drift tests before capital deployment (Vaver and Koehler 2011; Barajas and Bhamidipati 2021).

This study shifts causality to the start of the experimental pipeline, using Panel SEM to generate causal priors that are injected into DBSCAN stratification with constrained greedy matching to prevent cross-type contamination (Chen, Longfils, and Remy 2021). Moreover, this pre-deployment synthetic capital-injection and MDE validation can deliver decision-ready verification prior to allocating live capital, turning geo-experimentation from a retrospective measurement exercise into a proactive design tool.

Methodology

The methodology pipeline comprises four phases: (1) Causal Discovery via Structural Equation Modeling, (2) Geographic Stratification via Density-Based Clustering, (3) Cluster-Constrained Regional Matching, and (4) Minimum

Table 1: Methodological Comparison: Traditional Matching, Supergeo, and Causal-Informed Stratification

Feature / Methodology	Traditional Geo-Experiments	Modern Clustering (Supergeo / PCA)	Proposed Model
Regional Matching Logic	Greedy matching on pre-period revenue only (Chen, Longfils, and Remy 2021).	Mathematical balancing of pre-period covariates via MILP (Chen et al. 2023; Shaw 2025).	Cluster-Constrained Greedy Matching within causally identical cohorts.
Handling of Heterogeneity	Ignore—assumes all paired regions respond identically (Chen, Longfils, and Remy 2021).	Black-box variance minimization. Mathematically balances covariates via MILP, but the underlying structural homology cannot be explained or audited (Chen et al. 2023; Shaw 2025).	SEM priors provide a mathematical, auditable proof of causal alignment before matching occurs.
Dimensionality & Clustering	None (strict 1-to-1 pairings)	PCA (variance-based, uninterpretable, causally blind) + forced agglomeration (Chen et al. 2023; Shaw 2025).	Weighted DBSCAN—triages unmatchable outliers as "noise," preserving structural homology.
Parallel Trends Assumption	Highly vulnerable to drift during live test (Callaway and Sant’Anna 2021).	Mathematically balanced in the pre-period, but introduces unseen structural risk during live testing due to uninterpretable causal priors (Chen et al. 2023; Shaw 2025).	Highly Resilient and Auditable—matched regions share verified causal architecture, reacting symmetrically to external shocks.
Pre-Experiment Validation	Post-mortem only—risks launching underpowered tests (Barajas and Bhamidipati 2021).	Statistical boundary estimates, often lacking simulated geo-level capital injections (Shaw 2025; Chen et al. 2023).	Decision-Ready Pre-Validation—synthetic capital-injection calculates MDE, AA tests, and drift checks prior to capital deployment.

Detectable Effect (MDE) Validation. It is implemented in Python using semopy for structural modeling, scikit-learn for density-based clustering, and the Time-Based Regression (TBR) framework for the core regional match design and diagnostic validation (Chen, Longfils, and Remy 2021).

Dataset Description

The research data is based on a proprietary dataset that compiles daily performance indicators for localized capital deployments across 208 United States Designated Regional Areas (DMAs) (5,796 observations).

- **Capital Deployment Variables:** *cost_brand*, *cost_demandgen*, *cost_pmax*, & *cost_other*.
- **Behavioral Outcomes:** *total_visits* & *total_orders*.
- **Target KPI:** *revenue*.

All financial metrics represent scaled numerical values reflecting actual multi-channel capital deployment and resulting regional revenue. In the context of spatial risk modeling, these variables serve as an empirical proxy for diversified portfolio investments, representing distinct algorithmic (e.g., *cost_pmax*) and brand-driven (e.g., *cost_brand*) capital tranches. Intermediate behavioral outcomes (e.g., *total_visits* and *total_orders*) function as leading indicators of regional asset conversion, mediating the causal pathway between the initial capital injection and the realized revenue.

Causal Discovery via Structural Equation Modeling

To extract region-specific causal responsiveness priors, we estimate a panel Structural Equation Model (SEM) to establish the directed causal flow from controllable capital investments through behavioral outcomes to revenue.

Using the semopy framework on 5,796 observations, a Directed Acyclic Graph (DAG) is fit to isolate statistically significant pathways ($p < 0.05$). A Depth-First Search (DFS) algorithm traverses the DAG to determine the Total Effect of each investment lever on the target variable (Ngurah Satria Wijaya, Triandini, and Gabriela Kabnani 2020).

For a given investment variable v , let $\text{Direct}_{v \rightarrow \text{Revenue}}$ denote the direct path coefficient from v to revenue. Let $\text{Path}_{v \rightarrow \text{Revenue}}^{(k)}$ denote the k -th indirect path from v to revenue through one or more mediators (e.g., $v \rightarrow \text{visits} \rightarrow \text{revenue}$). The Total Effect TE_v is defined as:

$$TE_v = \text{Direct}_{v \rightarrow \text{Revenue}} + \sum_k \left(\prod_{e \in \text{Path}_k} \beta_e \right)$$

where β_e is the estimated coefficient along edge e in the SEM. The DFS algorithm recursively identifies all such paths and computes their product, summing them to obtain TE_v .

To ensure stability in the subsequent stratification phase, we normalize the raw Total Effects using a log-compression and min-max scaling procedure. The empirical weight W_v for variable v is computed as:

$$W_v = 0.1 + 0.9 \cdot \frac{\log(1 + |TE_v|) - \min_u \log(1 + |TE_u|)}{\max_u \log(1 + |TE_u|) - \min_u \log(1 + |TE_u|)}$$

where the target variable (revenue) is anchored to $W_{\text{revenue}} = 1.0$. This transformation compresses extreme coefficient values while preserving the relative causal ordering of the capital levers, preventing any single variable from dominating the subsequent DBSCAN clustering.

Geographic Stratification via Weighted DBSCAN

Following the derivation of causal weights, the 208 DMAs are grouped into structurally homogeneous cohorts. The daily time-series data is aggregated to the regional level, resulting in mean values for revenue, costs, and behavioral metrics for each region.

These aggregated features are standardized ($\mu=0$, $\sigma=1$) to account for scalar discrepancies between revenue and behavioral counts. Importantly, the standardized features are then multiplied by the causal responsiveness scores derived from the SEM.

Let $Z_{i,j}$ denote the standardized value of feature j for region i . Let W_j denote the SEM-derived causal weight for feature j . The weighted feature matrix $X_{i,j}$ used for DBSCAN clustering is computed as:

$$X_{i,j} = Z_{i,j} \times W_j$$

To ensure clustering is governed by causal topology rather than raw variance, we define a causally-weighted Pearson correlation distance metric, $d_{\text{corr}}(u, v)$, for any two regions u and v :

$$d_{\text{corr}}(u, v) = 1 - \frac{\sum_j (X_{u,j} - \bar{X}_u)(X_{v,j} - \bar{X}_v)}{\sqrt{\sum_j (X_{u,j} - \bar{X}_u)^2 \sum_j (X_{v,j} - \bar{X}_v)^2}}$$

DBSCAN ($\text{eps} = 0.10$, $\text{min_samples} = 10$) is applied using $d_{\text{corr}}(u, v)$. This dynamically partitions the geography into structurally homologous cohorts while explicitly isolating volatile, unmatchable regions as Noise (Cluster -1).

This weighting ensures that features with stronger causal influence on revenue (e.g., *cost_brand*) have greater impact on the correlation distance calculation, while features with weaker causal influence contribute proportionally less to the stratification.

The researchers use the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm with the parameters $\text{eps}=0.10$ and $\text{min_samples}=10$ (correlation distance metric). DBSCAN was explicitly selected over K-Means due to its ability to identify arbitrary structural shapes and effectively isolate highly volatile, structurally incompatible regions as Noise (Cluster -1) (Artioli, Maci, and Magri 2024). This prevents the extreme outliers from skewing the baseline definitions of the homogenous cohorts. From a capital allocation perspective, this level of isolation serves as an automatic risk-triage mechanism. In real finance operations, forced matching of extremely volatile, unmatchable regional assets introduces substantial unhedged variation. Hence, the model proposes that if a geographic unit cannot be structurally aligned into a homogeneous cohort, then strategic resources should be withheld from that region during experimental validation to avoid unquantified

financial vulnerability.

Cluster-Constrained Greedy Matching

Standard greedy matching across N geographic units requires $\mathcal{O}(N^2)$ distance computations, which is prohibitively expensive and myopic for large-scale DMAs. By constraining the search space to localized clusters of maximum size k ($k \ll N$), the computational complexity is reduced to $\mathcal{O}(C \cdot k^2)$, where C is the number of identified DBSCAN clusters. This constraint significantly improves scalability and ensures that the matching algorithm focuses on units with the highest structural similarity.

For each homogeneous cluster, it uses a greedy search with Time-Based Regression (TBR) causal impact matching to match potential treatment and control regions. To ensure sufficient control regions for synthetic control estimation, this search is capped at 20 treatment regions per matched design. Parameters enforce a minimum pre-period correlation ($R \geq 0.80$) and assess structural stability using three diagnostic safety gates.

- **AA Testing:** Empirically verifying pre-treatment baseline equilibrium between cohorts ($p > 0.10$). Making sure that before the period begins, the proposed test and control groups are balanced.
- **Brownian Bridge Tests:** Assessing the maximum cumulative deviation between control and treatment time-series to verify parallel trends.
- **Durbin-Watson Tests:** Verifying residual independence to reject severe auto-correlation, bounded strictly as $1.5 \leq DW \leq 2.5$.

Matches failing to satisfy all three structural diagnostics are flagged as invalid, ensuring that only robust experimental designs proceed to the simulation phase.

Minimum Detectable Effect (MDE) & Budget Feasibility

To validate statistical power before deploying capital, we simulate a synthetic scale-up (or hold-out) intervention. The historical baseline data is split into a pseudo-training period and a pseudo-test period (e.g., a 7-day test window). A synthetic investment signal—calculated from the user-defined budget and a target ROI threshold—is injected into the treatment geos during the pseudo-test period. The Time-Based Regression (TBR) model estimates the treatment effect from this synthetic data.

The Minimum Detectable Effect (MDE) is calculated as:

$$MDE = t_{1-\alpha/2, n-1} \cdot \sigma \cdot \sqrt{\frac{1 - R^2}{n}}$$

where:

- σ is the pre-period standard deviation of the treatment group's daily revenue,
- n is the number of pre-period observations,
- R is the correlation between the matched treatment and control geos,
- $t_{1-\alpha/2, n-1}$ is the critical value from the t -distribution for a two-sided test at $\alpha = 0.10$ (with $n - 1$ degrees of freedom).

This calculation targets a two-sided $\alpha = 0.10$ and 80% statistical power. A Signal-to-Noise Ratio is computed as $SNR = \text{Target Impact} / MDE$. If $SNR < 1.0$, the test is flagged as "Underfunded"; $1.0 \leq SNR < 3.0$ is "Moderate"; and $SNR \geq 3.0$ is "Strong", ensuring the experiment is adequately powered against regional structural volatility.

Results and Discussion

This chapter presents empirical findings from applying the automated causal framework to a 28-day dataset of 208 US Designated Market Areas (DMAs). Results are organized into the four methodological phases: causal discovery, geographic stratification, TBR design validation, and baseline comparison triangulating final intervention simulations.

Causal Discovery and Structural Responsiveness

Panel Structural Equation Modeling (SEM) established the causal pathways between capital deployment, behavioral metrics, and revenue, successfully pruning non-significant variables.

The analysis showed that media levers interact within an interconnected network, with `cost_brand` as the primary driver. Normalizing the total causal effects extracted via Depth-First Search yielded these empirical responsiveness weights: `weights_revenue` (1.0), `weights_cost_brand` (1.0), `weights_cost_demandgen` (0.81), `weights_cost_other` (0.76), `weights_cost_pmax` (0.56), `weights_total_orders` (0.34), and `weights_total_visits` (0.10).

These weights confirm that while `cost_brand` directly drives revenue, supporting channels like `cost_demandgen` and `cost_pmax` operate synergistically. They serve as quantitative priors for geographic stratification, grouping regions by structural responsiveness to localized capital interventions rather than just revenue size.

Geographic Stratification via Weighted DBSCAN

Projecting the 208 DMAs into a multi-dimensional space weighted by SEM priors enabled the weighted DBSCAN algorithm to identify latent structural boundaries, partitioning the geography into two homogeneous cohorts and isolating extreme outliers (Figure 1):

- **Cluster 0:** 162 DMAs (e.g., Charleston-Huntington WV, Fairbanks AK, Indianapolis IN); a homogeneous block of mid-to-high sensitivity regions.
- **Cluster 1:** 37 DMAs (e.g., New York NY, Detroit MI, Charlotte NC, Salt Lake City UT, Austin TX);
- **Noise (Outliers):** 9 DMAs; automatically isolated as structurally incompatible to prevent skewing the TBR matching algorithm.

Figure 1 shows DBSCAN stratification of 208 US DMAs into structurally homogeneous spatial cohorts. The axes represent the empirical testing variables: `cost_brand` serves as the primary capital investment tranche, while `avg_total_visits` acts as the regional behavioral proxy. Cluster 0 (162 DMAs) and Cluster 1 (37 DMAs) form dense, structurally aligned cohorts, while 9 volatile outliers are isolated as noise to prevent capital misallocation.

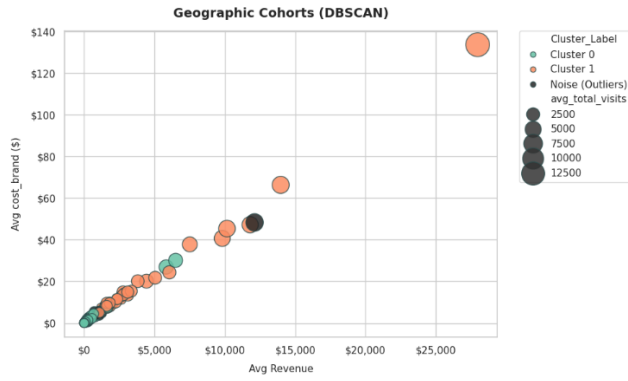


Figure 1: DBSCAN stratification of 208 US DMAs

For Cluster 0, a cluster-constrained greedy search identified 18 valid matched designs (correlations: 0.84–0.96). The optimal Match 1 (correlation: **0.9600**) was selected for power analysis. All matches passed AA, Brownian Bridge, and Durbin-Watson safety gates. For Cluster 1, 18 valid matches were identified (correlations: 0.92–0.98). The optimal Match 1 achieved a correlation of **0.9800**.

Pre-Experiment Validation and Power Analysis

The cluster-constrained region-matched design verified causal-informed stratification efficacy. By pairing control and treatment regions within assigned cohorts, the algorithm simulated a synthetic scale-up targeting a \$75,000 expected impact at a 3.0× target ROI.

Cluster 0 Validation

In Cluster 0, the optimal match passed all safety gates (AA Test, Drift Check, Durbin-Watson) with a pre-period correlation of $R = 0.9600$. The simulation yielded an MDE of **\$2,193 (34.2× signal-to-noise ratio)**, confirming a robust budget allocation immune to regional volatility.

Figure 2a visualizes this structural alignment ($R = 0.96$), while the unconstrained full-pool match (Figure 3) diverges significantly ($R = 0.65$), confirming the misalignment quantified in the baseline comparisons.

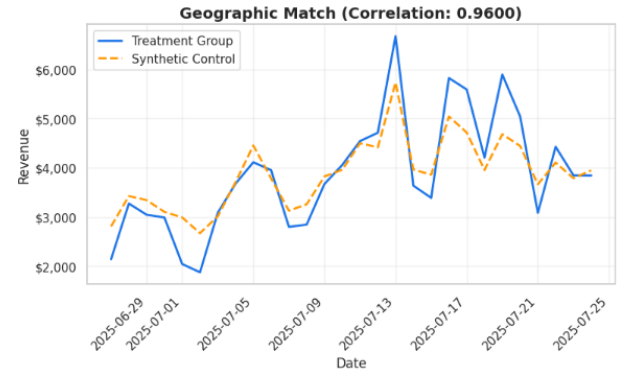
Cluster 1 Validation

For the high-volume metropolitan cohort, the optimal match passed safety gates ($R = 0.9800$). Its MDE of **\$15,232** achieved a **4.9× signal-to-noise ratio**, exceeding the 3.0× power threshold before deployment.

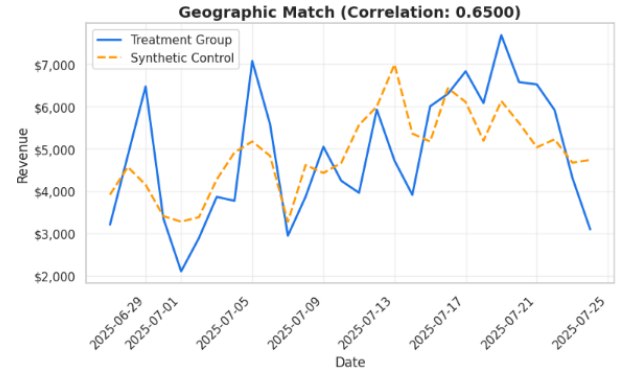
Pre-period trend verification on cluster-constrained matching Figure 2a produces a stable, reliable counterfactual ($R = 0.96$, all safety gates passed), while Unconstrained greedy matching Figure 2b on the full heterogeneous pool yields a weak, unreliable counterfactual ($R = 0.65$, Durbin-Watson Failed), visually confirming the necessity of SEM-informed stratification.

Constrained vs. Greedy Matching

To quantify the benefit of stratification, the authors ran the same matching algorithm on the full 208-DMA pool without



(a) Cluster 0 (Constrained): $R = 0.96$. Treatment and control move in near-perfect lockstep, validating structural alignment.



(b) Full Pool (Unconstrained): $R = 0.65$, DW Failed. Treatment and control exhibit significant divergence, indicating structural misalignment.

Figure 2: Pre-period trend verification comparing structurally constrained cohorts versus full-pool greedy matching.

clustering. The resulting match achieved a correlation of only **0.6500** and **failed the Durbin-Watson safety gate**, indicating severe autocorrelation and a structurally weak counterfactual (Figure 3).

Table 2: Baseline Comparisons: Matching Quality Comparison

Condition	Regions	Best Corr.	Safety Gates	SNR	Status
Full Pool (Unconstrained)	208	0.65	DW Failed	9.6×	Invalid
Cluster 0 (Constrained)	162	0.9600	All Passed	34.2×	Valid
Cluster 1 (Constrained)	37	0.9800	All Passed	4.9×	Valid

The Full Pool matching failed critically. While it produced a superficial SNR of 9.6x, the match achieved only $R = 0.65$ and explicitly failed the Durbin-Watson diagnostic, indicating severe autocorrelation. Any treatment effect estimates derived from this design would be irreparably bi-

ased. This explicitly demonstrates why observational algorithms (like PCA/Supergeo) fail: maximizing variance without causal boundaries forces the algorithm to pair structurally incompatible regions, breaking the parallel trends assumption. Cluster-constrained matching ensures that structural validity precedes statistical power.

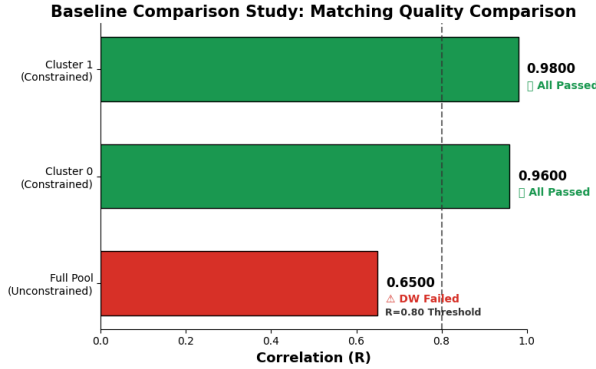


Figure 3: Baseline comparison of matching quality.

Figure 3 shows that unconstrained greedy matching on the full pool results to a structurally weak match ($R = 0.65$) and fails the Durbin-Watson safety gate, while cluster-constrained matching achieves consistently high correlations ($R \geq 0.96$) and passes all safety gates, validating the necessity of SEM-informed stratification.

Another noteworthy paradox emerges from this comparison as the Full Pool ($R=0.65$, DW Failed) produces a Signal-to-Noise Ratio of $9.6\times$, which would superficially suggest a "good" budget allocation. However, this is misleading—the design is structurally unsound due to autocorrelation. Any treatment effect estimate derived from this design would be biased, rendering its precise S/N ratio meaningless.

In contrast, Cluster 1 ($R=0.98$, All Passed) has a lower S/N ratio ($4.9\times$) but is structurally valid. The $4.9\times$ ratio still exceeds the 3.0 threshold, confirming it is sufficiently powered to detect the target treatment effect.

In addition, unbounded greedy algorithms can theoretically impose large pre-period connections by heavily weighting different regions (e.g. pairing a midwestern hub with a rural mountain hamlet). But such "Frankenstein" counterfactuals are based on artificial correlations. Without aligned causal priors (SEM responsiveness) these behaviors diverge in treatment, invalidating the parallel trends assumption and rendering treatment effect estimates fundamentally flawed.

These findings provide conclusive empirical evidence that **structural validity must precede statistical power**. The standard greedy algorithm produces structurally unsound counterfactuals, whereas the proposed framework enforces structural parity through SEM-informed stratification and rigorous safety gates.

Conclusion

The authors proposed a multi-stage causal framework for validating geographic experimental designs prior to capital deployment. The framework combines panel Structural

Equation Modeling (SEM) for causal pathway discovery, weighted DBSCAN for geographic stratification, and cluster-constrained greedy matching for structural alignment to the fundamental limitations of the standard "greedy" matching algorithms that match geographic regions based on superficial short-term revenue correlation only, rather than structural compatibility.

The empirical validation on 208 US Designated Market Areas (DMAs) confirms the efficacy of the proposed approach. The weighted DBSCAN algorithm geographic units into two dense cohorts, namely, Cluster 0 (162 DMAs) and Cluster 1 (37 DMAs), and isolates 9 volatile outliers as noise. Cluster-constrained matching within these cohorts consistently achieved high correlations ($R \geq 0.96$ for both clusters) and passed all safety gates (AA tests, Brownian Bridge, Durbin-Watson). The Minimum Detectable Effect (MDE) calculations yielded signal-to-noise ratios of $34.2\times$ for Cluster 0 and $4.9\times$ for Cluster 1, confirming that both designs are sufficiently powered to detect the target treatment effect.

The baseline comparison provides the strongest empirical evidence for the necessity of causal stratification. Unconstrained greedy matching on the full 208-DMA pool achieved only $R = 0.65$ and failed the Durbin-Watson safety gate, confirming that pure correlation-based matching produces structurally unsound counterfactuals. Furthermore, while modern variance-minimization techniques (e.g., Supergeo) may mathematically balance covariates in the short term, they operate as causal black boxes that force financial stakeholders to accept unquantifiable structural risks. This study demonstrates that enforcing SEM-informed stratification ensures both empirical structural validity and auditable causal homology precede statistical power—because a precise MDE is meaningless if the counterfactual is biased or uninterpretable.

The framework makes three contributions to applied computing and intelligent decision support: (1) an algorithmic pipeline integrating SEM, weighted DBSCAN, and cluster-constrained matching to reduce treatment effect heterogeneity; (2) a pre-experiment validation system that calculates MDE and conducts safety checks before capital deployment; and (3) empirical benchmarking through a controlled comparison demonstrating that causal-informed stratification is statistically necessary for reliable treatment effect measurement.

Limitations and Future Work

Three limitations should be acknowledged. First, the analysis is based on a 28-day pre-test window, which may not capture longer-term seasonality or trend dynamics. Second, the framework relies on fixed effects rather than external instruments for causal identification in the SEM; while E-value sensitivity analysis suggests robustness, this remains a limitation. Third, the 9 outlier DMAs identified as noise were excluded from the primary analysis; results may not generalize to structurally volatile geographic regions.

Future work will extend the framework to post-experiment treatment effect estimation using an ensemble of DiD, QRF, and Bayesian STS models and explore adaptive clustering for outlier markets. Longer pre-test windows and external

instrumental variables will also be investigated to strengthen causal identification.

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